



Course End Project Report

on

**Design a Horn Antenna that work in C-band
Microwave frequencies using HFSS**

A8423 – Antennas and Wave Propagation

Submitted by

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CONTENTS

S. No		Pg. No
1.	Introduction	
1.1.	Introduction to Antennas	3
1.2	CAD Tool – HFSS/MATLAB/FEKO/CST	4
1.3	Step involved in Design and Simulation	7
1.4	Objectives of the Project	8
2.	Antenna Design and Modelling	
2.1	Antenna Design	8
2.2	Antenna Geometrical Structure	9
2.3	Modelling of the Antenna in HFSS/MATLAB/FEKO/CST	9
3.	Results and Analysis	
3.1	Return Loss	10
3.2	Finalized Antenna Geometry and Return Loss	12
4.	Applications	12
5.	Program Outcomes & Sustainable Development Goals Mapping	13
6.	Conclusions	14

1. INTRODUCTION

1.1. Introduction to Antennas:

An antenna is a fundamental component in wireless communication systems, serving as a transducer that converts electrical signals into electromagnetic waves for transmission and vice versa for reception. Through this process, antennas enable the exchange of information over distances without the need for physical connections, making them indispensable in applications such as mobile communications, Wi-Fi, Bluetooth, etc.

The performance of an antenna is characterized by several key parameters:

Radiation pattern: This describes the distribution of radiated power as a function of direction, allowing identification of the antenna's principal radiating directions.

Gain: Antenna gain quantifies how efficiently input power is transformed into radio waves focused in a preferred direction, relative to a reference antenna.

Bandwidth: Bandwidth specifies the range of frequencies over which the antenna maintains satisfactory performance, ensuring reliable operation across desired channels.

Azimuth pattern: Represents the antenna radiation pattern in the horizontal plane (ϕ -plane), showing how the antenna radiates power around it at a fixed elevation angle, typically 0° .

Elevation pattern: Represents the radiation pattern in the vertical plane (θ -plane), illustrating how radiation varies with elevation angle at a fixed azimuth angle.

Return loss ($S(1,1)$): Return loss indicates the proportion of power reflected back toward the source due to impedance mismatch, with higher return loss values corresponding to better impedance matching and minimal reflected power.

Beamwidth: Beamwidth refers to the angular width of the antenna's main lobe, measured at the half-power (-3 dB) points, and defines the coverage area of the antenna's strongest signal.

These parameters collectively determine the suitability of an antenna for specific wireless applications, influencing its ability to transmit and receive signals effectively across intended frequency bands and spatial directions.

Introduction to Application involved in the Project assigned:

The C-band horn antenna operates in the 4 to 8 GHz frequency range and is widely utilized across several important microwave communication and sensing applications. The following table outlines common application areas and their relevance to the horn antenna design:

S.No	Application Area	Description
1	Satellite Communication	Used for uplink and downlink transmissions due to good atmospheric penetration and moderate antenna size.
2	Radar Systems	Provides directional high-gain beams essential for object detection and tracking in air and ground radar.
3	Microwave Links	Employed in point-to-point communication systems for reliable, high-frequency data transmission.
4	Scientific Instrumentation	Used in microwave measurement setups and electromagnetic wave experiments requiring specific beam patterns.

1.2. Introduction to HFSS :

ANSYS HFSS (High Frequency Structure Simulator) is a 3D electromagnetic simulation software used as a CAD tool for designing and analyzing RF and microwave components. It uses the Finite Element Method (FEM) to solve Maxwell's equations and accurately computes parameters like S-parameters, impedance, field distribution, and antenna radiation patterns. HFSS enables users to model complex high-frequency structures, apply boundary conditions and excitations, and visualize results through adaptive meshing for high accuracy, making it widely used in antenna design, microwave engineering, and high-speed electronic applications.

ANSYS (particularly ANSYS HFSS) includes a wide range of electromagnetic (EM) solvers based on different numerical techniques, each suited for specific types of problems:

- FEM (Finite Element Method):
Primary solver in HFSS for 3D full-wave EM simulation.
Best for high accuracy in complex geometries, resonant structures, waveguides, antennas, and filters.
Handles both open-boundary and closed structures.
- IE (Integral Equation Method):
Efficient for electrically large, open-region problems like scattering and radiation.
Reduces volume mesh requirements since it works with surface currents.

Often used in RCS (Radar Cross Section) analysis and antenna placement.

- PO (Physical Optics):
High-frequency asymptotic solver.
Best for very large structures where full-wave FEM/IE would be too computationally expensive.
Common in automotive radar, satellite communication, and reflector antennas.
- Hybrid Solutions (FEM-IE, FEM-PO, etc.):
Combine accuracy and efficiency.
Example: FEM models a detailed antenna feed, while PO handles the large reflector.
Great for multi-scale problems where small complex structures interact with large surfaces.

Antenna modelling in HFSS:

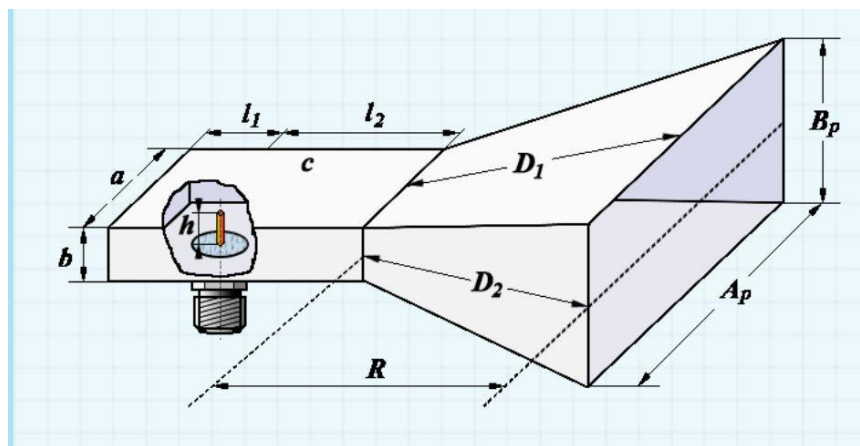


Fig-1: Design Structure of Horn antenna

Step 1: Start HFSS Project

- Open HFSS software.
- Insert a new project and select "Insert HFSS Design".

Step 2: Create the Cuboid

- Go to "Draw" and select "Box".
- Set the box center position: (0.795, -0.3975, 0.0)

- Dimensions:

XSize (width): 1.59 in (c)

YSize (height): 0.795 in (a)

ZSize (depth/thickness): 1 in (b)

- Rename the box appropriately (Waveguide).

Name	Value	Unit	Evaluated V...
Command	CreateRectangle		
Coordinate...	Global		
Position	-3.975,-1.9875,5	in	-3.975in,-1.9...
Axis	Z		
XSize	7.95	in	7.95in
YSize	3.975	in	3.975in

Fig-2: Dimensions of Cuboid

Step 3: Create the Rectangle

- Go to "Draw" and select "Rectangle".
- Set the rectangle position: (-3.975, -1.9875, 0.5)
- Dimensions:

XSize (flare end width): 7.95 in (Ap)

YSize (flare end height): 3.975 in (Bp)

- Rename the rectangle appropriately (HornFlare).

Name	Value	Unit	Evaluated V...
Command	CreateRectangle		
Coordinate...	Global		
Position	-3.975,-1.9875,5	in	-3.975in,-1.9...
Axis	Z		
XSize	7.95	in	7.95in
YSize	3.975	in	3.975in

Fig-3: Dimensions of Rectangle

Step 4: Unite all parts.

- Select both box and rectangle.
- Use Boolean operation to unite them into a single object.

Step 5: Create the Radiation boundary

- Go to "Draw" and select "Box".
- Set the box center position: (-4.4, -2.2, 0.0)
- Dimensions:

XSize (width): 8.8 mm

YSize (height): 4.4 mm

ZSize (depth/thickness): 15 mm

- Rename the box appropriately (RadiationBoundary).

Name	Value	Unit	Evaluated V...
Command	CreateBox		
Coordinate...	Global		
Position	-4.4,-2.2,0		
XSize	8.8	in	-4.4in,-2.2in...
YSize	4.4	in	4.4in
ZSize	15	in	15in

Fig-4: Dimensions of air box

Step 6: Final Design

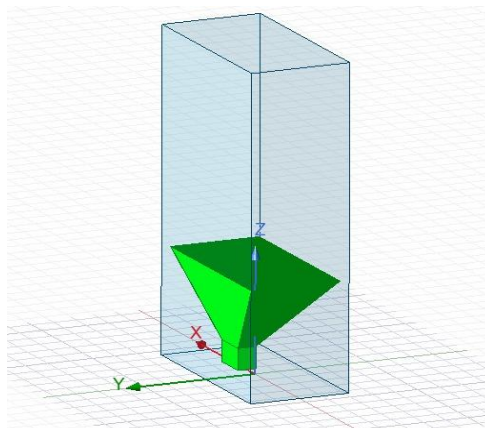


Fig-5: Design of Horn Antenna

1.3. Simulation Steps involved:

Step 1: Setup solution setup.

- Go to "Analysis Setup".
- Set center frequency around 6 GHz for C-band.
- Add frequency sweep from 4 GHz to 8 GHz covering full C-band.

Step 2: Save, validate, and analyze.

- Save the project after setup.
- Use "Validate" to check for errors.
- Run "Analyze All" to perform simulation.

Step 3: View results and generate plots:

- To plot **S-parameters (S11)**: Go to Results → Create Modal Solution Report → Rectangular Plot → Select S(1,1) → Choose solution and sweep → OK.
- Add marker on the lowest S11 point to find resonant frequency.
- To plot **Gain**: Results → Create Far Field Report → Rectangular Plot → Quantity: Gain → Select sweep → OK.
- To plot **Directivity**: Similar to Gain, select Directivity instead.

- To plot **VSWR**: Results → Create Modal Solution Report → Rectangular Plot → Quantity: VSWR → Select sweep → OK.
- To plot **Impedance (Z Parameter)**: Results → Create Modal Solution Report → Rectangular Plot → Quantity: Impedance real or imaginary → Select sweep → OK.
- To generate **3D Radiation Pattern**: Results → Create Far Field Report → 3D Polar Plot → Select solution and frequency → OK.

1.4. Objectives of the Project:

- Design a C-band horn antenna operating at 4 to 8 GHz.
- Model and simulate the antenna geometry using HFSS software.
- Analyze key parameters like gain, directivity, bandwidth, return loss, and VSWR.
- Optimize design for maximum efficiency and good impedance matching.
- Validate antenna performance for practical applications in satellite and radar systems.

2. Antenna Design and Modelling:

2.1. Antenna Design:

- Wavelength $\lambda = \frac{c}{f}$, where $c = 3 \times 10^8$ m/s (speed of light), $f = 6 \times 10^9$ Hz
 - Height of waveguide section $a = \frac{\lambda}{2}$
 - Depth or thickness $b \approx a$
 - Width of waveguide $c = \lambda a$
 - Aperture width $A_p = N \times a$
 - Aperture height $B_p = N \times b$
- c : speed of light
 f : operating frequency
 N : flare factor (typically between 5 and 10)

Fig-6: General formulae for Horn antenna

- Directivity: $D = \frac{4\pi U}{P_{rad}}$
- Gain: $G = e_{cd} D = e_{cd} \frac{4\pi U}{P_{rad}}$
- Decibel Gain: $G_{dB} = 10 \log_{10}(G)$

Geometry Parameters of Design:

Parameter	Online Calculator Value	Optimized Project Value
Height (a)	0.85 in	0.795 in
Depth (b)	1.1 in	1.0 in
Width (c)	1.7 in	1.59 in
Aperture Width (A_p)	8.2 in	7.95 in
Aperture Height (B_p)	4.1 in	3.975 in

2.2. Antenna Geometrical Structure:

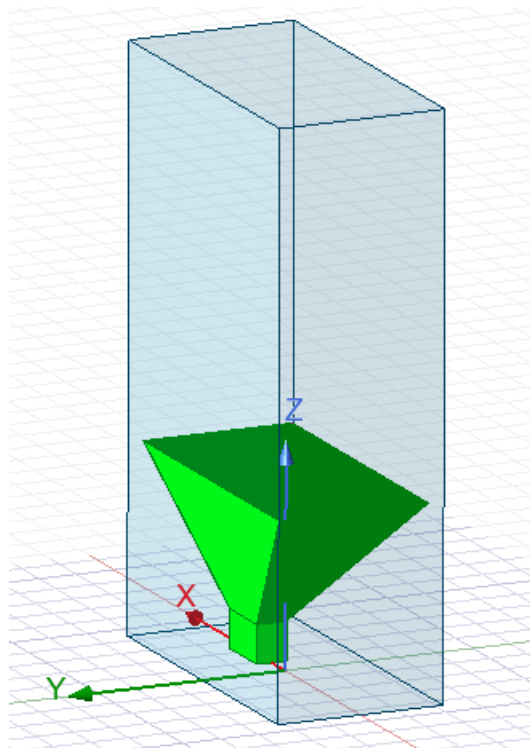


Fig-6: Design Structure of Horn antenna

2.3. Modelling of antenna in HFSS

In HFSS, the antenna structure is modelled by sequentially selecting 3D objects like rectangles, cylinders, or wires from the 'Draw' menu, placing them in the design space.

3. Results and Analysis

3.1. Return Loss:

S.No	Frequency (GHz)	Return Loss (dB)
1	6.01	-25.93

Observations:

1. S11 (Return Loss) Plot: Shows S11 in dB versus frequency. Dips indicate frequencies with good matching and minimal power reflection; for example, a deep notch at 6.01 GHz.

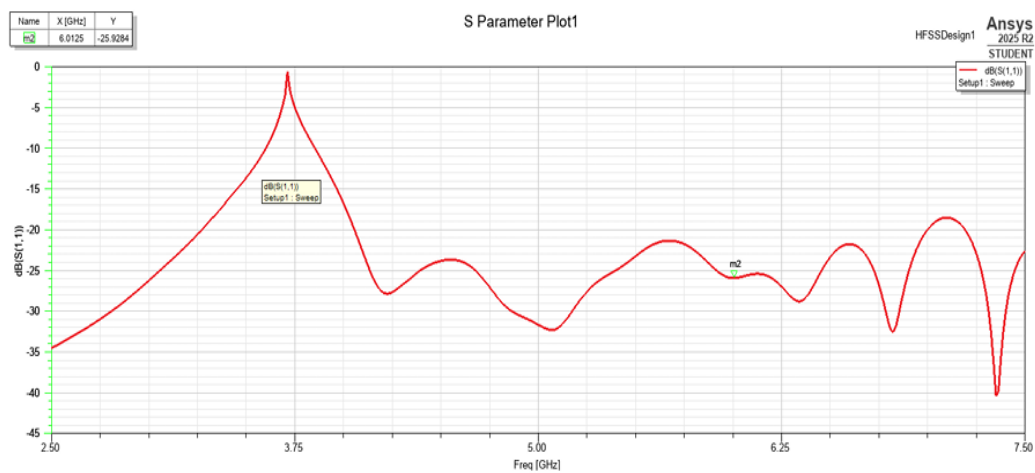


Fig-7: S11 Plot

2. VSWR Plot: Presents voltage standing wave ratio versus frequency. A value close to 1 indicates ideal matching and minimal signal reflection at the antenna port.

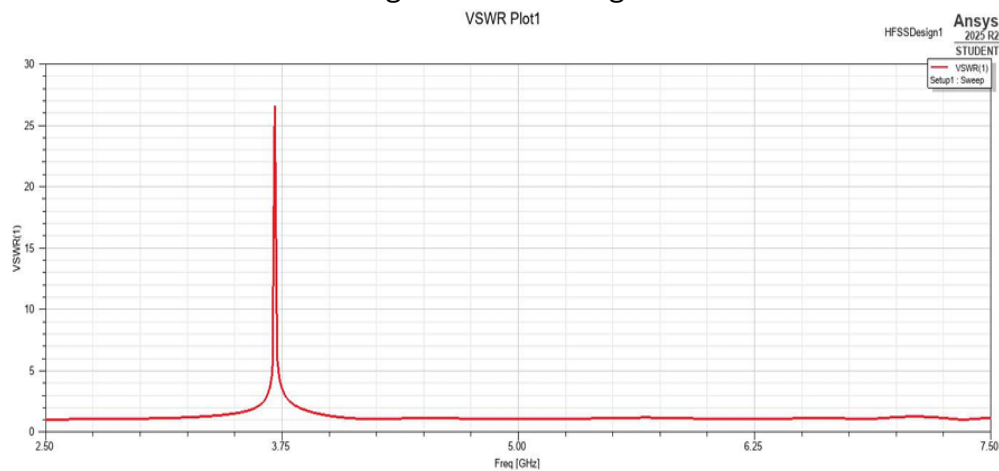
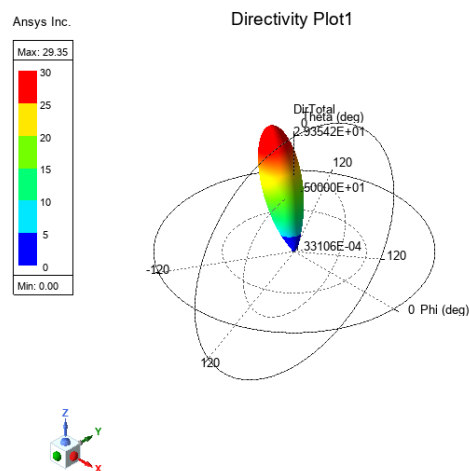


Fig-8: VSWR Plot

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4. Directivity Plot: Shows the total directivity of the horn antenna as a function of angle. The maximum directivity indicates how well the antenna focuses energy in a particular direction (main lobe)



11

5. Radiation Pattern Plot: 3D/2D plot illustrating how the antenna radiates energy in space at a specific frequency. It visually represents the main lobe, side lobes, and nulls of the antenna beam.

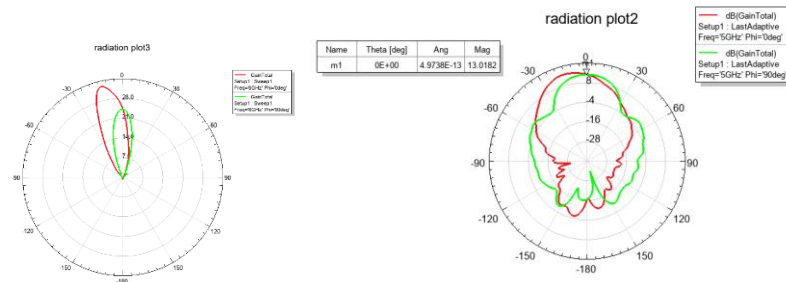


Fig-11: Radiation Pattern Plots

3.2. Finalized Antenna Geometry and Return Loss

The optimized design dimensions are smaller than the basic ideal dimensions calculated for 6 GHz, reflecting a compact design suitable for integration in tight spaces without compromising performance. From the return loss plot, the antenna shows excellent impedance matching at 6 GHz with a deep dip near -26 dB confirming efficient radiation. The aperture size and waveguide dimensions are sufficient to support the desired C-band frequencies while maintaining good directivity and gain.

4. Application:

Specification	Design Achievement	Application
Resonating Frequency	The designed antenna resonates at 6.01 GHz, within the C-band used for satellite and wireless systems.	Suitable for C-band satellite communication and radar systems operating around 6 GHz.
Return Loss and Gain	Achieved a return loss of approximately -25.93 dB, and gain values aligning with the desired antenna efficiency.	Ideal for high-performance wireless communication requiring low reflection and strong signal strength.
Beamwidth	Narrow beamwidth observed from radiation pattern plots enhancing directivity.	Used in radar, point-to-point communication, etc.

5.1. Project outcomes mapped to Program Outcomes:

Program outcome	Description	Justification
PO1	Engineering Knowledge	Applied electromagnetic theory and HFSS simulation to design and analyze horn antennas.
PO2	Problem Analysis	Analyzed antenna parameters like gain, return loss, and beamwidth to assess performance.
PO3	Design/Development of Solutions	Designed and optimized horn antenna models meeting specific wireless communication needs.
PO5	Modern Tool Usage	Used ANSYS HFSS effectively for simulation and performance evaluation of antenna design.
PSO2	Advanced Communication Systems Competency	Specialized in antenna design and analysis for advanced communication systems.

5.2. Project/Problem outcomes mapped to SDGs:

SDG Code	SDG Name	Justification
SDG 9	Industry, Innovation and Infrastructure	Supports innovation in wireless communication systems, aiding in the development of efficient communication infrastructures.
SDG 11	Sustainable Cities and Communities	Enables smart city applications with efficient communication networks, improving urban living through IoT and sensors.

6. Conclusions

This project successfully designed and simulated a C-band horn antenna using HFSS software. The antenna shows excellent performance with a return loss of about -26 dB at 6 GHz, indicating good impedance matching and efficient signal transmission. The gain and directivity results demonstrate that the antenna radiates well in the desired direction, suitable for modern wireless communication systems. Optimizing the horn dimensions helped achieve a compact design that meets the required specifications while maintaining high efficiency. Overall, this project provided valuable hands-on experience in antenna design, simulation, and analysis, laying a strong foundation for future work in RF communication technologies.